

Siyu Chen

☎ +1 2036276364 ✉ siyu.chen.sc3226@yale.edu 📍 New Haven, CT, USA 🐙 FFishy-git 🎓 Google Scholar

Research Interest

Deep Learning, Foundation Model, Various Aspects of Statistics, Reinforcement Learning

Research Highlights

Theoretical Foundations of Transformers & In-Context Learning

- **Unveiling Induction Heads: Provable Training Dynamics and Feature Learning in Transformers**
First end-to-end characterization of how a transformer with FFN learns general induction heads mechanisms — the attention circuit underlying in-context learning.
NeurIPS 2024
- **Training Dynamics of Multi-Head Softmax Attention for In-Context Learning: Emergence, Convergence, and Optimality**
First end-to-end characterization of the full training dynamics of multi-head softmax attention, proving convergence to an optimal in-context learning algorithm.
COLT 2024 (Oral) · ICLR 2024 Workshop on Bridging the Gap Between Practice and Theory in Deep Learning · **Invited Talk**, NYU Courant Math and Data Seminar · **Invited Talk**, Workshop on the Theory of Decision Intelligence, RLChina 2024
- *Both works above were presented in the **FAI Seminar**, 2024.*

Statistical & Computational Foundations of Learning

- **Can Neural Networks Achieve Optimal Computational-Statistical Tradeoff? An Analysis on Single-Index Model**
Proves that neural networks trained by gradient descent attain the optimal sample complexity for single-index models, matching statistical-query lower bounds.
ICLR 2025 (Oral)
- **An Optimized Franz–Parisi Criterion and Its Equivalence with SQ Lower Bounds**
Establishes an optimized Franz–Parisi criterion from statistical physics and proves its exact equivalence with statistical-query lower bounds for high-dimensional inference.
NeurIPS 2025 (Oral)

Self-evolution of LLMs

- **INFUSER: Influence-Guided Self-Evolution Improves Reasoning**
Introduces influence-guided self-evolution, where a co-evolving generator produces document-grounded curricula that improve solver reasoning.
Preprint, 2026

Education

Doctoral Student

Yale University, Statistics and Data Science

Sept. 2022 to now

- *Coursework*: Linear models, Statistical Inference, Practical Work, Data Analysis, Intermediate Machine Learning, Advanced Optimization
- *Advisor*: Prof. Zhuoran Yang

Bachelor of Science	Tsinghua University , Electronic Engineering • <i>Coursework</i>: Convex Optimization, Computer Program Design, Linear Algebra, Probability and Stochastic Processes, Theory and Methods for Statistical Inference • <i>Honors & Awards</i>: – National Second Prize for Contemporary Undergraduate Mathematical Contest in Modeling – Comprehensive Excellence Scholarship – Jiang Nanxiang Award – China National Scholarship	Sept. 2018 to May 2022
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Experience

Yale University , Teaching Assistant	CT, USA
• Information Theory (S&DS 364)	Feb 2024 to May 2024
• Optimization and Computation (S&DS 431)	Sep 2024 to Dec 2024
• Deep Learning for Scientists and Engineers (S&DS 266)	Feb 2025 to May 2025
Google Research , Student researcher (CPT)	CT, USA
• Research on the test-time scaling behavior and workflow optimization for large language model.	June 2025 to Dec 2025 6 months

Publications

- Altabaa, A., S. Chen, J. Lafferty, and Z. Yang (2025). “Unlocking Out-of-Distribution Generalization in Transformers via Recursive Latent Space Reasoning”. In: *arXiv preprint arXiv:2510.14095*.
- Chen, S., M. Lu, B. Wu, H. Sheen, F. Zhang, S. Li, Z. Li, J. Blanchet, T. Wang, and Z. Yang (2026). “INFUSER: Influence-Guided Self-Evolution Improves Reasoning”. In: *arXiv preprint arXiv:2606.09052*.
- Chen, S., T. Misiakiewicz, I. Zadik, and P. Zhang (2026). “An optimized franz-parisi criterion and its equivalence with sq lower bounds”. In: *Advances in Neural Information Processing Systems* 38. **Oral**, pp. 110461–110506.
- Chen, S., H. Sheen, T. Wang, and Z. Yang (2024). “Unveiling Induction Heads: Provable Training Dynamics and Feature Learning in Transformers”. In: *Advances in Neural Information Processing Systems*. Vol. 37, pp. 66479–66567.
- Chen, S., H. Sheen, X. Xiong, T. Wang, and Z. Yang (2026). “Taming Polysemanticity in LLMs: Theory-Grounded Feature Recovery via Sparse Autoencoders”. In: *The Fourteenth International Conference on Learning Representations*.
- Chen, S., M. Wang, and Z. Yang (2023). “Actions Speak What You Want: Provably Sample-Efficient Reinforcement Learning of the Quantal Stackelberg Equilibrium from Strategic Feedbacks”. In: *arXiv preprint arXiv:2307.14085*.
- Chen, S., Y. Wang, Z. Wang, and Z. Yang (2023). “A unified framework of policy learning for contextual bandit with confounding bias and missing observations”. In: *arXiv preprint arXiv:2303.11187*.
- Chen, S., B. Wu, M. Lu, Z. Yang, and T. Wang (2025). “Can neural networks achieve optimal computational-statistical tradeoff? an analysis on single-index model”. In: *International Conference on Learning Representations*. Vol. 2025. **Oral**, pp. 71176–71260.
- Chen, S., J. Wu, Y. Wu, and Z. Yang (2023). “Learning to incentivize information acquisition: Proper scoring rules meet principal-agent model”. In: *International Conference on Machine Learning*. PMLR, pp. 5194–5218.
- Chen, S., D. Yang, J. Li, S. Wang, Z. Yang, and Z. Wang (2022). “Adaptive model design for Markov decision process”. In: *International Conference on Machine Learning*. PMLR, pp. 3679–3700.
- He, J., S. Chen, F. Zhang, and Z. Yang (2024). “From Words to Actions: Unveiling the Theoretical Underpinnings of LLM-Driven Autonomous Systems”. In: *International Conference on Machine Learning*. PMLR, pp. 17807–17841.

- He, J., X. Pan, S. Chen, and Z. Yang (2025). “In-Context Linear Regression Demystified: Training Dynamics and Mechanistic Interpretability of Multi-Head Softmax Attention”. In: *International Conference on Machine Learning*. PMLR, pp. 22686–22742.
- He, J., L. Wang, S. Chen, and Z. Yang (2026). “On the Mechanism and Dynamics of Modular Addition: Fourier Features, Lottery Ticket, and Grokking”. In: *arXiv preprint arXiv:2602.16849*.
- Hu, X., F. Zhang, S. Chen, and Z. Yang (2024). “Unveiling the Statistical Foundations of Chain-of-Thought Prompting Methods”. In: *arXiv preprint arXiv:2408.14511*.
- Sheen, H., S. Chen, T. Wang, and H. H. Zhou (2024). “Implicit Regularization of Gradient Flow on One-Layer Softmax Attention”. In: *arXiv preprint arXiv:2403.08699*.
- Siyu, C., S. Heejune, W. Tianhao, and Y. Zhuoran (2024). “Training Dynamics of Multi-Head Softmax Attention for In-Context Learning: Emergence, Convergence, and Optimality”. In: *The Thirty Seventh Annual Conference on Learning Theory*. PMLR, pp. 4573–4573.
- Wu, J., S. Chen, M. Wang, H. Wang, and H. Xu (2024). “Contractual Reinforcement Learning: Pulling Arms with Invisible Hands”. In: *arXiv preprint arXiv:2407.01458*.
- Zhang*, Y., S. Chen*, Z. Yang, M. Jordan, and Z. Wang (2021). “Wasserstein flow meets replicator dynamics: A mean-field analysis of representation learning in actor-critic”. In: *Advances in Neural Information Processing Systems 34*, pp. 15993–16006.

Technologies

Languages: Python, C/C++, R, Bash

LLM Development: PyTorch, Hugging Face Transformers, vLLM/sglang, Megatron, slime, verl

Training & Evaluation: RLVR, prompt engineering, benchmark evaluation